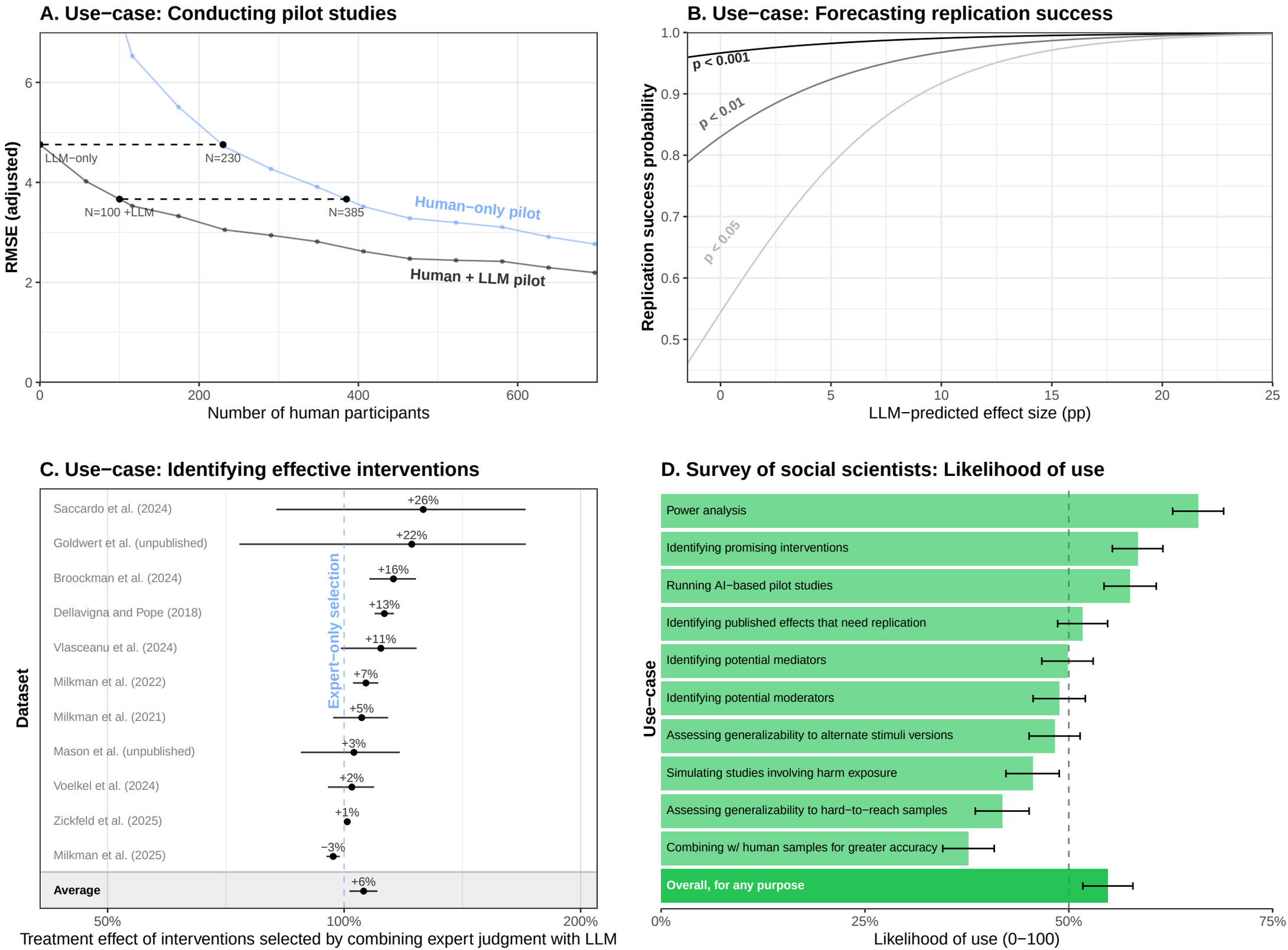
A. Use−case: Conducting pilot studies
RMSE (adjusted)
Number of human participants
LLM−only
N=230
N=100 +LLM
N=385
Human−only pilot
Human + LLM pilot
B. Use−case: Forecasting replication success
Replication success probability
LLM−predicted effect size (pp)
p < 0.001
p < 0.01
p < 0.05
C. Use−case: Identifying effective interventions
Dataset
Treatment effect of interventions selected by combining expert judgment with LLM
Expert−only selection
Saccardo et al. (2024) +26%
Goldwert et al. (unpublished) +22%
Broockman et al. (2024) +16%
Dellavigna and Pope (2018) +13%
Vlasceanu et al. (2024) +11%
Milkman et al. (2022) +7%
Milkman et al. (2021) +5%
Mason et al. (unpublished) +3%
Voelkel et al. (2024) +2%
Zickfeld et al. (2025) +1%
Milkman et al. (2025) −3%
Average +6%
D. Survey of social scientists: Likelihood of use
Use−case
Likelihood of use (0−100)
Power analysis
Identifying promising interventions
Running AI−based pilot studies
Identifying published effects that need replication
Identifying potential mediators
Identifying potential moderators
Assessing generalizability to alternate stimuli versions
Simulating studies involving harm exposure
Assessing generalizability to hard−to−reach samples
Combining w/ human samples for greater accuracy
Overall, for any purpose